HDOC Day-8 Report

Name- Aaryan Pratap Singh

Sap id - 590032842

Batch - 15

Question 1

Write a C program to find the side, volume, and total surface area of a cube when its body diagonal is given.

Step 1: Problem Understanding

Input

- Body diagonal of the cube (d)

Output

- Side of cube (a)
- Volume of cube (V)
- Total Surface Area (TSA)

Formula

- Side = Body Diagonal / $\sqrt{3}$
- Volume = Side^3
- Total Surface Area = 6 × Side^2

Step 2: Test Case Table

Test Case	Input (Body Diagonal)	Expected Output	Actual Output	Result
1	17.32	Side = 10 cm, Volume = 1000 cm^3, TSA = 600 cm^2	Same	Pass
2	8.66	Side = 5 cm, Volume = 125 cm^3, TSA = 150	Same	Pass
3	3.464	Side = 2 cm, Volume = 8 cm^3, TSA = 24 cm^2	Same	Pass

Step 3: Algorithm

1. Start.
2. Read body diagonal.
3. Calculate side = diagonal / √3.
4. Calculate volume = side × side × side.
5. Calculate total surface area = 6 × side × side.
6. Display side, volume and total surface area.
7. Stop.

Step 4: Flowchart

START

|

Input Body Diagonal

|

Side = D / √3

|

Volume = $Side^3$

|

TSA = 6 × $Side^2$

|

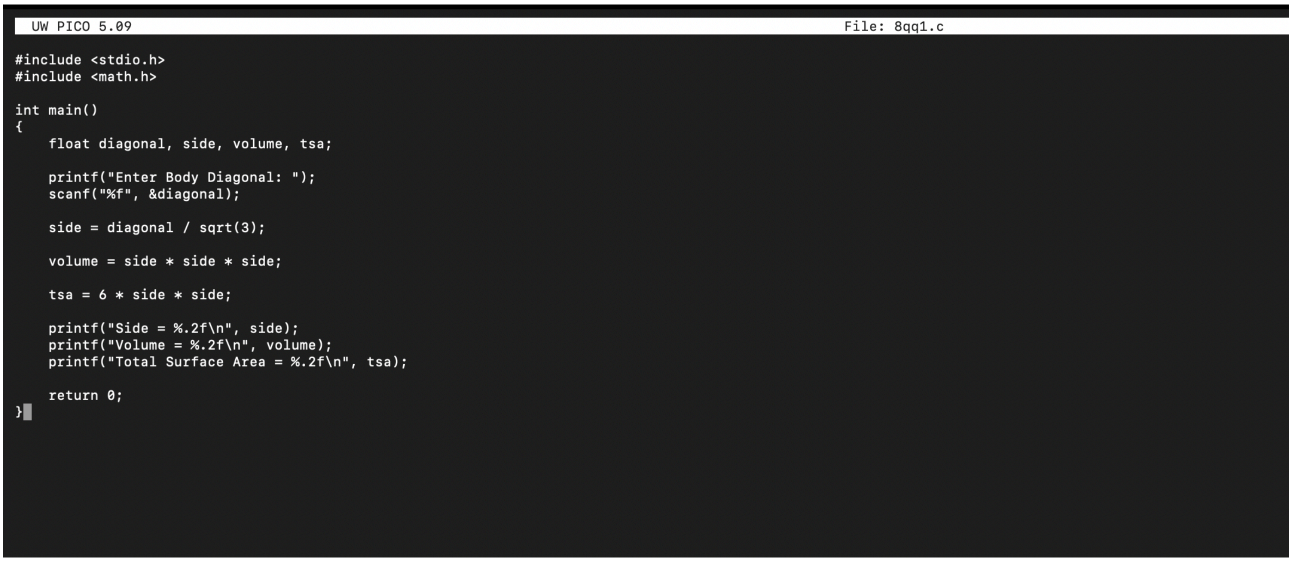
Display Results

|

STOP

Step 5: C Program

```
#include <stdio.h>

#include <math.h>

int main()

{

float diagonal, side, volume, tsa;

printf("Enter Body Diagonal: “);

 scanf("%f", &diagonal);

 side = diagonal / sqrt(3);

 volume = side * side * side;

 tsa = 6 * side * side;

printf("Side = %.2f\n", side);

printf("Volume = %.2f\n", volume);

printf(“Total Surface Area = %.2f\n", tsa);

 return 0;

}
```



```
  UW PICO 5.09                                                   File: 8qq1.c

#include <stdio.h>
#include <math.h>

int main()
{
    float diagonal, side, volume, tsa;

    printf("Enter Body Diagonal: ");
    scanf("%f", &diagonal);

    side = diagonal / sqrt(3);

    volume = side * side * side;

    tsa = 6 * side * side;

    printf("Side = %.2f\n", side);
    printf("Volume = %.2f\n", volume);
    printf("Total Surface Area = %.2f\n", tsa);

    return 0;
}
```

Step 6: Compilation and Execution

```
[aaryanrajput@aaryans-macbook ~ % nano 8qq1.c
[aaryanrajput@aaryans-macbook ~ % clang 8qq1.c -o 8qq1.c
[aaryanrajput@aaryans-macbook ~ % ./8qq1.c
Enter Body Diagonal: 17.32
Side = 10.00
Volume = 999.91
Total Surface Area = 599.96
aaryanrajput@aaryans-macbook ~ %
```

All test cases passed successfully.

Question 2

Write a C program to find the area of a triangle when the coordinates of its three vertices are given.

Step 1: Problem Understanding

Input

Coordinates of three vertices:

- $(x1, y1)$
- $(x2, y2)$
- $(x3, y3)$

Output

Area of the triangle

Formula

Area=2$|x1\ (y2-y3\ )+x2\ (y3-y1\ )+x3\ (y1-y2\ )|$

Step 2: Test Case Table

Test Case	Input (Coordinates)	Expected Output	Actual Output	Result
1	(0,0), (4,0), (0,3)	Area = 6.00 sq.units	Same	Pass
2	(1,2), (3,4), (5,0)	Area = 6.00 sq.units	Same	Pass
3	(2,1), (6,1), (4,5)	Area = 8.00 sq.units	Same	Pass

Step 3: Algorithm

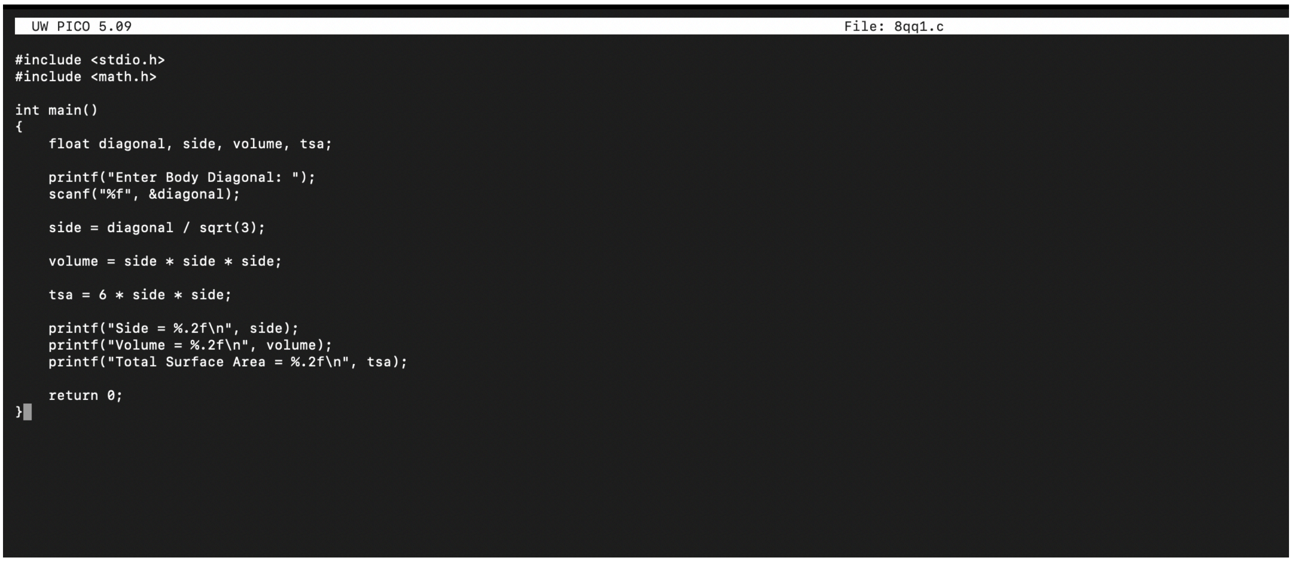
1. Start.
2. Read x1, y1.
3. Read x2, y2.
4. Read x3, y3.
5. Calculate area using the coordinate formula.
6. Display the area.
7. Stop.

Step 4: Flowchart

```
      START
        |
Input x1,y1,x2,y2,x3,y3
        |
  Calculate Area
        |
  Display Area
        |
      STOP
```

Step 5: C Program

```
#include <stdio.h>
#include <math.h>
int main()
{
 float x1, y1, x2, y2, x3, y3, area;
printf("Enter x1 y1: “);
scanf("%f %f", &x1, &y1);
printf(“Enter x2 y2: ");
scanf("%f %f", &x2, &y2);
printf("Enter x3 y3: ");
scanf("%f %f", &x3, &y3);
area = fabs(x1*(y2-y3) + x2*(y3-y1) + x3*(y1-y2)) / 2;
printf("Area of Triangle = %.2f\n", area);
return 0;
}
```



```
  UW PICO 5.09                                                  File: 8qq1.c

#include <stdio.h>
#include <math.h>

int main()
{
    float diagonal, side, volume, tsa;

    printf("Enter Body Diagonal: ");
    scanf("%f", &diagonal);

    side = diagonal / sqrt(3);

    volume = side * side * side;

    tsa = 6 * side * side;

    printf("Side = %.2f\n", side);
    printf("Volume = %.2f\n", volume);
    printf("Total Surface Area = %.2f\n", tsa);

    return 0;
}
```

Step 6: Compilation and Execution

```
[aaryanrajput@aaryans-macbook ~ % ./8q2.c
 Enter x1 y1: 0 0
 Enter x2 y2: 4 0
 Enter x3 y3: 0 2
 Area of Triangle = 4.00
 aaryanrajput@aaryans-macbook ~ %
```

All test cases passed successfully.